

A Compact Assignment-Based Reformulation for Transmission-Network Reduction

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Abstract—This note considers assignment-based reduction of a meshed transmission network. The original model moves bus injections and line endpoints to retained buses, but contains products between binary assignments and continuous voltage angles. The reformulation replaces these products with line-status variables, lifted angles, physical reduced-network flows, and an auxiliary internal-redistribution flow. The two formulations are equivalent when every cluster is connected: physical transfer between clusters is carried by f , injection relocation inside clusters by g^{int} . Because the clustering is itself a decision variable, the bound switching g^{int} on and off cannot depend on the clusters; a bound computable from the input data alone is derived. A multi-objective version trading reduction against worst-case flow error is given in the appendices.

Index Terms—Transmission-network reduction, clustering, DC power flow, mixed-integer linear programming, multi-objective optimization.

I. ORIGINAL PROBLEM

Consider a connected transmission network with N buses and L lines. The data are

$$p \in \mathbb{R}^N, \quad \hat{\theta} \in \mathbb{R}^N, \quad D^x \in \mathbb{R}^{L \times L}, \quad E \in \mathbb{R}^{N \times L}, \quad (1)$$

where p is the nodal injection vector, $\hat{\theta}$ the original voltage-angle vector, D^x the diagonal line-susceptance matrix, and E the node–line incidence matrix. The original DC power flow is

$$p = ED^xE^T\hat{\theta}, \quad (2)$$

$$\hat{f} = D^xE^T\hat{\theta}. \quad (3)$$

The binary matrix $A \in \{0, 1\}^{N \times N}$ defines the reduction:

$$A_{ij} = 1 \iff \text{bus } j \text{ is assigned to retained bus } i. \quad (4)$$

Thus Ap moves injections and AE moves line endpoints. Each bus is assigned once, and only a retained bus can receive assignments:

$$\sum_{i=1}^N A_{ij} = 1, \quad \forall j, \quad (5)$$

$$A_{ij} \leq A_{ii}, \quad \forall i, j. \quad (6)$$

The objective $\min \text{tr}(A)$ minimizes the number of retained buses. Every cluster must induce a connected subgraph.

For line l with endpoints u_l, v_l : the line is *internal* if both endpoints share a retained bus, and *external* otherwise. Internal lines disappear after contraction; external lines remain. Write $C_i := \{j : A_{ij} = 1\}$ and $\|x\|_1 := \sum_j |x_j|$.

Remark 1 (Connectivity constraints). A_{conn} is any linear formulation whose feasible A are exactly those making each

C_i connected. A standard choice is a single-commodity flow model with binary arc variables $y_l \leq c_l$ selecting a spanning forest, each retained bus a root, each assigned bus consuming one unit, and arc capacities Ny_l . Connectivity is assumed throughout and underlies both directions of the equivalence proof.

II. ORIGINAL FORMULATION

Let $\theta \in \mathbb{R}^N$ contain candidate retained-bus angles. The angle assigned to every original bus is $A^T\theta$, so the reduced flow is

$$f = D^xE^TA^T\theta, \quad (7)$$

and the assigned power balance is

$$Ap = AEF = AED^xE^TA^T\theta. \quad (8)$$

Let $\underline{f}_l, \bar{f}_l$ be the physical limits and ϵ_l^T the allowed error relative to $\hat{f}_l$. These apply only to external lines:

$$|f_l - \hat{f}_l| \leq \epsilon_l^T, \quad \forall l \text{ external}, \quad (9)$$

$$\underline{f}_l \leq f_l \leq \bar{f}_l, \quad \forall l \text{ external}. \quad (10)$$

An absorbed line has equal endpoint angles, so (7) gives $f_l = 0$. Internal lines are therefore deliberately exempt from (9)–(10): contraction removes the line from the reduced network, and the power it carried is not lost but reappears as internal redistribution, represented below by g^{int} .

Define the originally binding lines

$$\mathcal{L}^+ := \{l : \hat{f}_l = \bar{f}_l\}, \quad \mathcal{L}^- := \{l : \hat{f}_l = \underline{f}_l\}. \quad (11)$$

To keep an existing congestion from disappearing through aggregation, every line in $\mathcal{L}^+ \cup \mathcal{L}^-$ is required to stay external, with

$$f_l \geq \hat{f}_l, \quad \forall l \in \mathcal{L}^+, \quad (12)$$

$$f_l \leq \hat{f}_l, \quad \forall l \in \mathcal{L}^-. \quad (13)$$

The model is exact but difficult: (7) and (8) both contain products of A with continuous angles.

III. REFORMULATION

A. Which Lines Are Internal?

Introduce $c_l \in \{0, 1\}$ with $c_l = 1$ iff u_l and v_l share a cluster, enforced by

$$c_l \geq A_{i,u_l} + A_{i,v_l} - 1, \quad \forall i, l, \quad (14)$$

$$c_l \leq 1 - A_{i,u_l} + A_{i,v_l}, \quad \forall i, l, \quad (15)$$

$$c_l \leq 1 + A_{i,u_l} - A_{i,v_l}, \quad \forall i, l. \quad (16)$$

Binding lines stay visible:

$$c_l = 0, \quad \forall l \in \mathcal{L}^+ \cup \mathcal{L}^-. \quad (17)$$

B. Physical Reduced-Network Flow

Introduce a lifted angle vector $\vartheta \in \mathbb{R}^N$ and define

$$\boxed{f = D^x E^\top \vartheta}, \quad f_l = D_{ll}^x (\vartheta_{u_l} - \vartheta_{v_l}). \quad (18)$$

This is the DC line-flow equation and the only place the susceptances act. It is not a mere definition of f : eliminating ϑ confines f to the range of $D^x E^\top$, a subspace of dimension $N - 1$. In a meshed network with $L > N - 1$ this imposes $L - N + 1$ independent restrictions, one per independent cycle $\mathcal{K}$,

$$\sum_{l \in \mathcal{K}} \sigma_l x_l f_l = 0, \quad x_l := 1/D_{ll}^x, \quad (19)$$

with $\sigma_l = \pm 1$ orienting l around the cycle. This is Kirchhoff's voltage law: flows cannot be chosen independently inside their windows, and the reactance ratios decide which combinations are admissible. Contracting a cluster deletes its internal lines from (19), which is the sense in which an absorbed line acts as an infinite-susceptance connection.

Define the allowed external-line interval

$$\underline{\phi}_l := \begin{cases} \max\{\underline{f}_l, \hat{f}_l - \epsilon_l^L, \hat{f}_l\}, & l \in \mathcal{L}^+, \\ \max\{\underline{f}_l, \hat{f}_l - \epsilon_l^L\}, & \text{otherwise,} \end{cases} \quad (20)$$

$$\bar{\phi}_l := \begin{cases} \min\{\bar{f}_l, \hat{f}_l + \epsilon_l^L, \hat{f}_l\}, & l \in \mathcal{L}^-, \\ \min\{\bar{f}_l, \hat{f}_l + \epsilon_l^L\}, & \text{otherwise,} \end{cases} \quad (21)$$

and impose

$$\boxed{\phi_l(1 - c_l) \leq f_l \leq \bar{\phi}_l(1 - c_l)} \quad \forall l, \quad (22)$$

which has two cases only: $c_l = 0$ gives $\underline{\phi}_l \leq f_l \leq \bar{\phi}_l$ (external), and $c_l = 1$ gives $f_l = 0$ (internal).

Remark 2 (Binding lines are reproduced exactly). For $l \in \mathcal{L}^+$, $\hat{f}_l = \bar{f}_l$, so (20)–(21) give $\underline{\phi}_l = \bar{\phi}_l = \hat{f}_l$: the window collapses to a point and ϵ_l^L has no effect there. Likewise for $\mathcal{L}^-$. This is the intended consequence of combining (10) with (12)–(13). If it is too strong, replace $\hat{f}_l$ in (12)–(13) by $\hat{f}_l \mp \delta_l$ for small $\delta_l \geq 0$.

Since $D_{ll}^x \neq 0$, $f_l = 0$ on an internal line forces $\vartheta_{u_l} = \vartheta_{v_l}$. Cluster connectivity then makes ϑ constant on each cluster, so

$$\vartheta = A^\top \theta \quad (23)$$

for some θ . This is obtained automatically, without binary–continuous products.

C. Injection Relocation Inside a Cluster

The flow f balances the reduced network only at cluster level, not at every eliminated bus. Introduce $g^{\text{int}} \in \mathbb{R}^L$, an auxiliary transfer used only inside clusters; it is not a physical DC flow and obeys no susceptance–angle relation. Impose

$$\boxed{p = E(f + g^{\text{int}})}, \quad (24)$$

so f carries physical power between clusters and g^{int} redistributes injections inside them. The support of g^{int} is enforced by

$$\boxed{-G c_l \leq g_l^{\text{int}} \leq G c_l} \quad \forall l, \quad (25)$$

with G constructed in Section IV: external lines get $g_l^{\text{int}} = 0$, internal lines get $|g_l^{\text{int}}| \leq G$.

Constraint (25) cannot be dropped. Every column of E has one $+1$ and one -1 , so $\mathbf{1}^\top E = 0$; with $\mathbf{1}^\top p = 0$ this makes $p - Ef$ zero-sum for every f , hence (24) solvable for every f . Unrestricted g^{int} therefore makes the balance vacuous, $c \equiv 1$ and $f \equiv 0$ feasible, and the network contracts to one bus.

Combining (18) and (24),

$$\boxed{p = ED^x E^\top \vartheta + Eg^{\text{int}}}, \quad (26)$$

so ϑ must produce susceptance-dependent flows satisfying both balance and the external-line bounds. Fix one reference angle,

$$\vartheta_{j_0} = 0. \quad (27)$$

Every constraint sees ϑ only through $E^\top \vartheta$, so this is a gauge choice: it removes the uniform-angle freedom without changing the feasible flows.

IV. A VALID BOUND ON THE INTERNAL TRANSFER

The clustering is a decision variable, so G cannot be built from the clusters: it must hold simultaneously for every A the model may select. The bound below is computed from the input data before solving and is independent of A . Define

$$F_m := \max\{|\underline{\phi}_m|, |\bar{\phi}_m|\}, \quad (28)$$

an upper bound on $|f_m|$ in either case of (22), since $c_m = 1$ gives $f_m = 0 \leq F_m$.

Lemma 1 (Data-only bound). *Let*

$$\boxed{G := \frac{1}{2}\|p\|_1 + \sum_{m=1}^L F_m}. \quad (29)$$

Then for every A satisfying (5), (6) and $A \in \mathcal{A}_{\text{conn}}$, and every feasible original solution, there exists g^{int} supported on internal lines with $r = Eg^{\text{int}}$ and $|g_l^{\text{int}}| \leq G$ for all l .

Proof: Let $r := p - Ef$; Section V shows $Ar = 0$, i.e. $\sum_{j \in \mathcal{C}_i} r_j = 0$ for every cluster.

Step 1 (route on a spanning tree). Each $\mathcal{C}_i$ is connected, so it admits a spanning tree $\mathcal{T}_i$. Set $g_l^{\text{int}} = 0$ off $\bigcup_i \mathcal{T}_i$. On a tree, $r = Eg^{\text{int}}$ restricted to $\mathcal{C}_i$ has a unique solution: removing $l \in \mathcal{T}_i$ splits $\mathcal{C}_i$ into $\mathcal{S}_l$ and $\mathcal{C}_i \setminus \mathcal{S}_l$, and orienting l out of $\mathcal{S}_l$,

$$g_l^{\text{int}} = \sum_{j \in \mathcal{S}_l} r_j. \quad (30)$$

Step 2 (take the smaller side). The residual is zero-sum on the cluster, so $\sum_{j \in \mathcal{S}_l} r_j = -\sum_{j \in \mathcal{C}_i \setminus \mathcal{S}_l} r_j$ and

$$|g_l^{\text{int}}| \leq \frac{1}{2} \sum_{j \in \mathcal{C}_i} |r_j| \leq \frac{1}{2} \|r\|_1. \quad (31)$$

Here the cluster leaves the argument: any cluster is a subset of the bus set, so the whole network is the worst case.

Step 3 (bound the residual by data). $\|r\|_1 \leq \|p\|_1 + \|Ef\|_1$, and each column of E has two nonzeros, so

$$\|Ef\|_1 \leq \sum_j \sum_m |E_{jm}| |f_m| = 2 \sum_m |f_m| \leq 2 \sum_m F_m. \quad (32)$$

With (31), $|g_l^{\text{int}}| \leq \frac{1}{2}\|p\|_1 + \sum_m F_m = G$. ■

Remark 3 (One-sided). Any $G' \geq G$ is valid, so overestimating only weakens the linear relaxation. Underestimating removes feasible clusterings and turns the model into a restriction of the original rather than an equivalent form.

Remark 4 (Tightening under a cardinality cap). If $\mathcal{A}_{\text{conn}}$ also enforces $|C_i| \leq K$, only buses in the cluster and lines touching it contribute to (31). With Δ the maximum bus degree, a cluster touches at most $K\Delta$ lines, so

$$G(K) := \frac{1}{2} \sum_{j \in \mathcal{K}} |p_j| + \sum_{m \in \mathcal{M}} F_m \quad (33)$$

is valid, where $\mathcal{K}$ indexes the K largest $|p_j|$ and $\mathcal{M}$ the $K\Delta$ largest F_m . This is usually far smaller than (29) and still data-only.

Remark 5 (Avoiding the constant). Solvers supporting indicator constraints can replace (25) by $c_l = 0 \Rightarrow g_l^{\text{int}} = 0$, needing no constant and typically branching better. Lemma 1 is then required only for reporting that g^{int} is bounded.

Remark 6 (A spanning forest suffices). Step 1 uses only tree lines, so g^{int} never needs support on more than $N - \text{tr}(A)$ lines. If $\mathcal{A}_{\text{conn}}$ supplies forest-arc variables $y_l \leq c_l$ (Remark 1), replacing c_l by y_l in (25) cuts the switched constraints from L to the forest size and permits the tighter bound (33).

V. WHY THE REFORMULATION IS EQUIVALENT

Assume every cluster is connected and G satisfies Lemma 1.

A. Original Model Implies Reformulation

From a feasible original (A, θ) , set

$$\vartheta = A^T \theta - (A^T \theta)_{j_0} \mathbf{1}. \quad (34)$$

A shift by a multiple of $\mathbf{1}$ is invisible to E^T , so (18) reproduces the original flow (7) while (27) holds. Internal lines have equal endpoint angles and zero flow, satisfying (22) with $c_l = 1$; external lines satisfy the same window in both models, satisfying (22) with $c_l = 0$.

With $r := p - Ef$, the original balance (8) gives $Ar = 0$: the entries of r sum to zero inside every cluster. Lemma 1 then supplies a g^{int} supported on internal lines with $|g_l^{\text{int}}| \leq G$, so (24) and (25) hold.

B. Reformulation Implies Original Model

From a feasible reformulated solution, internal lines have $f_l = 0$, hence equal endpoint angles by (18), hence (23) for some θ by connectivity. Multiplying (24) by A ,

$$A(p - Ef) = AEg^{\text{int}} = 0, \quad (35)$$

because an internal line has a zero column in AE (both endpoints share an assignment) and an external line has $g_l^{\text{int}} = 0$ by (25). Therefore $Ap = AEf$, and substituting $f = D^x E^T \vartheta$ with $\vartheta = A^T \theta$ recovers (8).

The flow constraints match line by line: for $c_l = 0$, (22) is exactly (9)–(10) with (12)–(13); for $c_l = 1$, both models impose $f_l = 0$ and neither imposes a window. The two models have the same feasible clusterings and the same optimal value, with angles agreeing up to the gauge (34).

VI. COMPLETE REFORMULATED PROBLEM

Let $\mathcal{A}_{\text{conn}}$ denote the selected connected-clustering constraints. The compact formulation is

$$\min_{A, c, \vartheta, f, g^{\text{int}}} \text{tr}(A) \quad (36a)$$

$$\text{s.t.} \quad \sum_i A_{ij} = 1, \quad \forall j, \quad (36b)$$

$$A_{ij} \leq A_{ii}, \quad \forall i, j, \quad (36c)$$

$$A \in \mathcal{A}_{\text{conn}}, \quad A_{ij} \in \{0, 1\}, \quad \forall i, j, \quad (36d)$$

$$c_l \in \{0, 1\}, \quad \forall l, \quad (36e)$$

$$(14) \text{--}(16), \quad \forall i, l, \quad (36f)$$

$$c_l = 0, \quad \forall l \in \mathcal{L}^+ \cup \mathcal{L}^-, \quad (36g)$$

$$f = D^x E^T \vartheta, \quad (36h)$$

$$\phi_l(1 - c_l) \leq f_l \leq \bar{\phi}_l(1 - c_l), \quad \forall l, \quad (36i)$$

$$p = E(f + g^{\text{int}}), \quad (36j)$$

$$-G c_l \leq g_l^{\text{int}} \leq G c_l, \quad \forall l, \quad (36k)$$

$$\vartheta_{j_0} = 0. \quad (36l)$$

The continuous variables have distinct roles: ϑ are reduced-network bus angles, f are physical susceptance-dependent line flows, and g^{int} is redistribution inside clusters only. The reformulation removes the $O(N^2)$ assignment-angle products, replacing them with line-level binary constraints and bounded internal transfers.

Susceptances act only through (18): retained lines keep their original values and no equivalent admittances are created, in contrast with Ward-type reduction. The resulting mismatch is absorbed by ϵ^L and by the relocation g^{int} . Making susceptances variable would reintroduce bilinear terms and is excluded.

VII. TRADING REDUCTION AGAINST FLOW ERROR

In (36) the tolerance ϵ^L is fixed and the reduction is optimized. The two conflict directly: a looser tolerance admits more contraction. Treating both as objectives produces the whole trade-off curve.

A. Making the Error a Variable

Let $\varepsilon \geq 0$ be the worst-case external-line flow error. The window (20)–(21) depends on ϵ_l^L and must be split. Keep the tolerance-free part,

$$\underline{\psi}_l := \begin{cases} \hat{f}_l, & l \in \mathcal{L}^+, \\ \underline{f}_l, & \text{otherwise,} \end{cases} \quad \bar{\psi}_l := \begin{cases} \hat{f}_l, & l \in \mathcal{L}^-, \\ \bar{f}_l, & \text{otherwise,} \end{cases} \quad (37)$$

replace (22) by the physical-limit switch

$$\underline{\psi}_l(1 - c_l) \leq f_l \leq \bar{\psi}_l(1 - c_l), \quad \forall l, \quad (38)$$

and add the error-measuring pair

$$f_l - \hat{f}_l \leq \varepsilon + |\hat{f}_l| c_l, \quad \forall l, \quad (39)$$

$$\hat{f}_l - f_l \leq \varepsilon + |\hat{f}_l| c_l, \quad \forall l. \quad (40)$$

The constant $|\hat{f}_l|$ is the point of the construction. For an external line, $c_l = 0$ and (39)–(40) reduce to $|f_l - \hat{f}_l| \leq \varepsilon$. For

an internal line, $c_l = 1$ and $f_l = 0$, so both read $|\hat{f}_l| \leq \varepsilon + |\hat{f}_l|$, true for every $\varepsilon \geq 0$: the line is released, as intended. Since ε may be zero, $|\hat{f}_l|$ is the smallest constant with this property, and it is read directly from (3).

Two consequences. First, F_m in (28) must be recomputed from the wider window, $F_m = \max\{|\underline{\psi}_m|, |\overline{\psi}_m|\}$, before evaluating G . Second, if line ratings differ widely, divide (39)–(40) by a scale $s_l > 0$ (e.g. $s_l = \bar{f}_l$) so ε measures relative error and is not dominated by the largest line.

APPENDIX A BI-OBJECTIVE FORMULATION

Both objectives are minimized; G uses $F_m = \max\{|\underline{\psi}_m|, |\overline{\psi}_m|\}$ in (29).

B. Objectives and How to Trace the Curve

Both $J_1 := \text{tr}(A)$ (retained buses) and $J_2 := \varepsilon$ (worst-case error) are minimized. The full bi-objective model is Appendix A.

Since J_1 is integer-valued in $\{1, \dots, N\}$, the Pareto front has at most N points and is a staircase. The efficient route is therefore to fix the bus budget and minimize the error: for $\beta = 1, \dots, N$ solve the single-objective model of Appendix B, giving $\varepsilon^*(\beta)$. Enlarging the budget can never hurt, so ε^* is non-increasing and the points $(\beta, \varepsilon^*(\beta))$ are exactly the Pareto front; monotonicity permits bisection or warm starting from $\beta + 1$.

Two alternatives are weaker. Fixing $\bar{\varepsilon}$ and minimizing $\text{tr}(A)$ recovers (36) when $\bar{\varepsilon}$ matches a uniform ϵ^L , but the grid of $\bar{\varepsilon}$ must be guessed, whereas the budget sweep is finite and exhaustive. Minimizing $\text{tr}(A) + \lambda \varepsilon$ is a single MILP, but the front is nonconvex because J_1 is integer, so unsupported Pareto points are unreachable for every λ , and λ must reconcile a bus count with a power quantity.

The curve $\varepsilon^*(\beta)$ is the practical output: the smallest achievable worst-case flow error for each reduced-network size. Its knee marks where further contraction starts to cost accuracy quickly. Lines in $\mathcal{L}^+ \cup \mathcal{L}^-$ stay external by (17) and contribute zero error by Remark 2, so ε^* is driven by the non-binding lines.

$$\min_{A, c, \vartheta, f, g^{\text{int}}, \varepsilon} (\text{tr}(A), \varepsilon) \quad (41a)$$

$$\text{s.t.} \quad \sum_i A_{ij} = 1, \quad \forall j, \quad (41b)$$

$$A_{ij} \leq A_{ii}, \quad \forall i, j, \quad (41c)$$

$$A \in \mathcal{A}_{\text{conn}}, \quad A_{ij} \in \{0, 1\}, \quad \forall i, j, \quad (41d)$$

$$c_l \in \{0, 1\}, \quad \forall l, \quad (41e)$$

$$c_l \geq A_{i, u_l} + A_{i, v_l} - 1, \quad \forall i, l, \quad (41f)$$

$$c_l \leq 1 - A_{i, u_l} + A_{i, v_l}, \quad \forall i, l, \quad (41g)$$

$$c_l \leq 1 + A_{i, u_l} - A_{i, v_l}, \quad \forall i, l, \quad (41h)$$

$$c_l = 0, \quad \forall l \in \mathcal{L}^+ \cup \mathcal{L}^-, \quad (41i)$$

$$f = D^x E^\top \vartheta, \quad (41j)$$

$$\underline{\psi}_l(1 - c_l) \leq f_l \leq \overline{\psi}_l(1 - c_l), \quad \forall l, \quad (41k)$$

$$f_l - \hat{f}_l \leq \varepsilon + |\hat{f}_l| c_l, \quad \forall l, \quad (41l)$$

$$\hat{f}_l - f_l \leq \varepsilon + |\hat{f}_l| c_l, \quad \forall l, \quad (41m)$$

$$p = E(f + g^{\text{int}}), \quad (41n)$$

$$-G c_l \leq g_l^{\text{int}} \leq G c_l, \quad \forall l, \quad (41o)$$

$$\vartheta_{j_0} = 0, \quad \varepsilon \geq 0. \quad (41p)$$

Efficient solutions are obtained by scalarization; the budget form below is the recommended one.

APPENDIX B MINIMUM WORST-CASE ERROR UNDER A BUS BUDGET

Given an integer budget $\beta \in \{1, \dots, N\}$ on retained buses, $\varepsilon^*(\beta)$ solves

$$\min_{A, c, \vartheta, f, g^{\text{int}}, \varepsilon} \varepsilon \quad (42a)$$

$$\text{s.t. } \text{tr}(A) \leq \beta, \quad (42b)$$

$$\sum_i A_{ij} = 1, \quad \forall j, \quad (42c)$$

$$A_{ij} \leq A_{ii}, \quad \forall i, j, \quad (42d)$$

$$A \in \mathcal{A}_{\text{conn}}, \quad A_{ij} \in \{0, 1\}, \quad \forall i, j, \quad (42e)$$

$$c_l \in \{0, 1\}, \quad \forall l, \quad (42f)$$

$$c_l \geq A_{i, u_l} + A_{i, v_l} - 1, \quad \forall i, l, \quad (42g)$$

$$c_l \leq 1 - A_{i, u_l} + A_{i, v_l}, \quad \forall i, l, \quad (42h)$$

$$c_l \leq 1 + A_{i, u_l} - A_{i, v_l}, \quad \forall i, l, \quad (42i)$$

$$c_l = 0, \quad \forall l \in \mathcal{L}^+ \cup \mathcal{L}^-, \quad (42j)$$

$$f = D^x E^T \vartheta, \quad (42k)$$

$$\underline{\psi}_l(1 - c_l) \leq f_l \leq \bar{\psi}_l(1 - c_l), \quad \forall l, \quad (42l)$$

$$f_l - \hat{f}_l \leq \varepsilon + |\hat{f}_l| c_l, \quad \forall l, \quad (42m)$$

$$\hat{f}_l - f_l \leq \varepsilon + |\hat{f}_l| c_l, \quad \forall l, \quad (42n)$$

$$p = E(f + g^{\text{int}}), \quad (42o)$$

$$-G c_l \leq g_l^{\text{int}} \leq G c_l, \quad \forall l, \quad (42p)$$

$$\vartheta_{j_0} = 0, \quad \varepsilon \geq 0. \quad (42q)$$

Three notes. The objective rewards retaining buses, so (42b) is tight at optimum and writing it as $\text{tr}(A) = \beta$ gives the same ε^* ; the inequality is preferred because it leaves the branch-and-bound tree less constrained. Feasibility for $\beta = N$ is immediate ($A = I$, $c = 0$, $g^{\text{int}} = 0$, $\vartheta = \hat{\theta} - \hat{\theta}_{j_0} \mathbf{1}$, $\varepsilon = 0$), and infeasibility for small β is caused by (17) together with connectivity, not by the error variable. Finally, $\varepsilon^*(\beta)$ is non-increasing in β , so sweeping β downward from N and warm starting each solve from the previous one traces the Pareto front in at most N solves.

APPENDIX C EDGE-BASED REFORMULATION AND TIGHT INTERNAL-TRANSFER BOUNDS

The assignment matrix can be removed by representing the reduction directly through the line-status variables

$$c_l = \begin{cases} 1, & \text{line } l \text{ is internal,} \\ 0, & \text{line } l \text{ is external.} \end{cases}$$

The clusters are recovered after optimization as the connected components of the subgraph induced by $\{l : c_l = 1\}$.

Let $f \in \mathbb{R}^L$ denote the physical DC line flow and $g \in \mathbb{R}^L$ the auxiliary redistribution flow supported only on internal lines. The proposed formulation is

$$\max_{c, \vartheta, f, g} \sum_{l=1}^L c_l \quad (43a)$$

$$\text{s.t. } c_l \in \{0, 1\}, \quad \forall l, \quad (43b)$$

$$c_l = 0, \quad \forall l \in \mathcal{L}^+ \cup \mathcal{L}^-, \quad (43c)$$

$$f = D^x E^T \vartheta, \quad (43d)$$

$$\underline{\phi}_l(1 - c_l) \leq f_l \leq \bar{\phi}_l(1 - c_l), \quad \forall l, \quad (43e)$$

$$p = E(f + g), \quad (43f)$$

$$-\underline{G}_l c_l \leq g_l \leq \bar{G}_l c_l, \quad \forall l, \quad (43g)$$

$$\vartheta_{j_0} = 0. \quad (43h)$$

If $c_l = 0$, then $g_l = 0$, and line l carries the physical flow

$$f_l = D_{ll}^x (\vartheta_{u_l} - \vartheta_{v_l})$$

inside its admissible window. If $c_l = 1$, then (43e) gives $f_l = 0$, and hence

$$\vartheta_{u_l} = \vartheta_{v_l},$$

while g_l may redistribute power inside the cluster. Thus, only (43g) requires a big- M or indicator implementation.

The objective in (43a) minimizes the number of external lines, not necessarily the number of retained buses. At optimality, every non-protected line whose endpoints are already connected by an internal path is also declared internal: its endpoint angles are equal, hence its physical flow is already zero, and changing c_l from zero to one preserves feasibility while improving the objective.

A. Tightening the Internal-Transfer Bound

The internal flow satisfies

$$Eg = p - Ef.$$

Let $\hat{f}$ be the original DC flow and define the original balance mismatch

$$e^0 := p - E\hat{f}. \quad (44)$$

Normally, $e^0 = 0$. Define the flow deviation

$$d := \hat{f} - f.$$

Then

$$p - Ef = e^0 + Ed. \quad (45)$$

For a line that may become internal, f_l belongs to

$$\{0\} \cup [\underline{\phi}_l, \bar{\phi}_l].$$

Therefore, a valid interval for d_l is

$$\underline{d}_l = \min \left\{ \hat{f}_l, \hat{f}_l - \bar{\phi}_l, \hat{f}_l - \underline{\phi}_l \right\}, \quad (46)$$

$$\bar{d}_l = \max \left\{ \hat{f}_l, \hat{f}_l - \bar{\phi}_l, \hat{f}_l - \underline{\phi}_l \right\}. \quad (47)$$

For a line fixed external, the internal case is removed:

$$\underline{d}_l = \hat{f}_l - \bar{\phi}_l, \quad \bar{d}_l = \hat{f}_l - \underline{\phi}_l. \quad (48)$$

Using (45), valid nodal residual bounds are

$$\underline{r}_i = e_i^0 + \sum_{l=1}^L \min\{E_{il}\underline{d}_l, E_{il}\bar{d}_l\}, \quad (49)$$

$$\bar{r}_i = e_i^0 + \sum_{l=1}^L \max\{E_{il}\underline{d}_l, E_{il}\bar{d}_l\}. \quad (50)$$

Define

$$R_i^+ := \max\{0, \bar{r}_i\}, \quad R_i^- := \max\{0, -\underline{r}_i\}. \quad (51)$$

For every feasible cluster, the residual sums to zero. Routing the residual on a spanning tree of the cluster gives, for every used internal edge,

$$|g_l| \leq \min \left\{ \sum_{i=1}^N R_i^+, \sum_{i=1}^N R_i^- \right\}.$$

Hence, the following data-only bound is valid:

$$G_{\text{res}} := \min \left\{ \sum_{i=1}^N R_i^+, \sum_{i=1}^N R_i^- \right\}. \quad (52)$$

A simpler alternative is obtained by defining

$$\Delta_l := \max\{|\underline{d}_l|, |\bar{d}_l|\}.$$

Since

$$\|e^0 + Ed\|_1 \leq \|e^0\|_1 + 2 \sum_{l=1}^L \Delta_l,$$

the bound

$$G_{\Delta} := \frac{1}{2} \|e^0\|_1 + \sum_{l=1}^L \Delta_l \quad (53)$$

is also valid.

The recommended common bound is

$$G = \min \left\{ G_{\text{res}}, G_{\Delta}, \frac{1}{2} \|p\|_1 + \sum_{l=1}^L F_l \right\}, \quad (54)$$

$$F_l := \max\{|\underline{\phi}_l|, |\bar{\phi}_l|\}. \quad (55)$$

Then set

$$\underline{G}_l = \bar{G}_l = G$$

in (43g). The new bounds are typically much tighter because they depend on the admissible deviations from the original flow, rather than on the full injection and line-flow magnitudes.